

Predicting Online News Popularity Using Machine Learning Methods

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MAIN GOAL:

In today's digital world, the success of online news platforms depends heavily on their ability to understand and predict what type of content will be attractive for readers. Accurate popularity prediction helps media companies optimize their content strategy by identifying which topics, writing styles, and sentiment patterns are most engaging to readers.

The main goal of our project is to identify the key factors that influence News popularity and to develop a model that predicts the popularity level of a future article based on these identified factors.

The main questions we aim to answer are: which specific factors have the greatest influence on the popularity of a given article (we will assess popularity by examining the target variable **shares** — the number of views), and how we can predict the popularity level of an article using the existing data.

DATA PREPROCESSING

In this stage of the project we have reviewed the data with 39,797 online news articles (Fernandes et al., 2015). Before the application of classification model, we have conducted the preprocessing of data. The names of the variables in the dataset were modified so they don't contain spaces. We have checked the data for any missing values and duplicates and made sure that the data is ready to be analyzed. We have analyzed the main parameters of the target variable "shares" to get a better understanding of the structure of the data.

EXPLORATORY DATA ANALYSIS

After the preprocessing of the data we began to explore the data deeper. Online News Popularity data base includes a huge number of features, most of which have little to do with the popularity of a news article (Figure 1). Reviewing the correlation of all variables with the target variable we saw that "shares" variable correlates the strongest with "kw_avg_avg" variable that represents the average popularity (in shares) of keywords that appear in the article. However, even the strongest correlation that we could find is just 0.11, which is such a small correlation that we would usually neglect it. This tells us that there is no feature that would correlate with target variable significantly, therefore, the popularity of a news article is hard to predict.

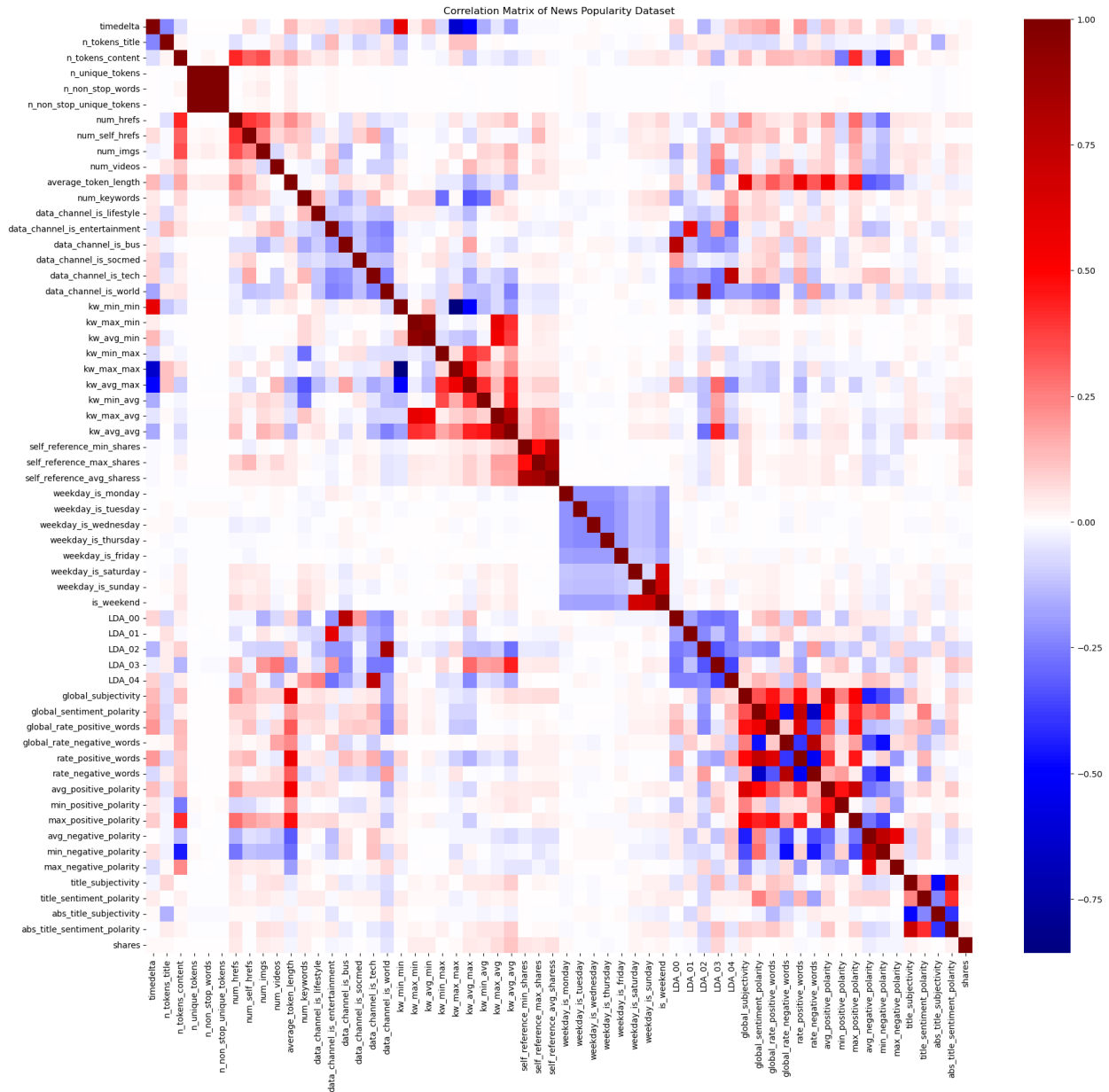


Figure 1: Correlation Heatmap of features for Online News Popularity data

CLASSIFICATION MODEL

To classify with articles will be popular we have identified a popularity threshold. In the exploratory data analysis stage, we found out that target variable values are skewed, there are many outliers in the data: news articles that differ significantly in popularity from the general mass. While this data is important to keep in the data base, it makes the mean value not representative of the actual average shares. Because of that we chose median to be a popularity threshold, that will distinguish popular articles from non-popular.

As we saw previously, original features from the dataset don't have much influence on the target variables. We have created 5 new variables: "virality_score", "topic_specialization", "text_complexity", "sentiment_power", and "seo_strength" using the ones we had in the data originally. We wanted to evaluate whether the newly engineered features contain predictive power for the target variable "popular" and whether any of them alone can significantly improve classification performance. For each of the features individually we have trained a Random Forest classifier using GridSearchCV:

Evaluating feature: virality_score

Best parameters: {'max_depth': 5, 'max_features': 'sqrt', 'n_estimators': 150}
 Best CV accuracy: 0.54
 Test accuracy: 0.54

Classification report:				
	precision	recall	f1-score	support
0	0.534	0.729	0.616	4017
1	0.555	0.346	0.426	3912
accuracy			0.540	7929
macro avg	0.544	0.538	0.521	7929
weighted avg	0.544	0.540	0.522	7929

Evaluating feature: topic_specialization

Best parameters: {'max_depth': None, 'max_features': 'sqrt', 'n_estimators': 150}
 Best CV accuracy: 0.609
 Test accuracy: 0.603

Classification report:				
	precision	recall	f1-score	support
0	0.600	0.650	0.624	4017
1	0.607	0.555	0.580	3912
accuracy			0.603	7929
macro avg	0.604	0.603	0.602	7929
weighted avg	0.604	0.603	0.602	7929

Evaluating feature: text_complexity

Best parameters: {'max_depth': 5, 'max_features': 'sqrt', 'n_estimators': 300}
 Best CV accuracy: 0.52
 Test accuracy: 0.526

Classification report:				
	precision	recall	f1-score	support
0	0.528	0.601	0.562	4017
1	0.522	0.448	0.483	3912
accuracy			0.526	7929
macro avg	0.525	0.525	0.522	7929
weighted avg	0.525	0.526	0.523	7929

Evaluating feature: sentiment_power

Best parameters: {'max_depth': 5, 'max_features': 'sqrt', 'n_estimators': 150}
 Best CV accuracy: 0.539
 Test accuracy: 0.542

Classification report:				
	precision	recall	f1-score	support
0	0.564	0.421	0.482	4017
1	0.528	0.665	0.589	3912
accuracy			0.542	7929
macro avg	0.546	0.543	0.535	7929
weighted avg	0.546	0.542	0.535	7929

Evaluating feature: seo_strength

Best parameters: {'max_depth': 5, 'max_features': 'sqrt', 'n_estimators': 150}
 Best CV accuracy: 0.525
 Test accuracy: 0.529

Classification report:				
	precision	recall	f1-score	support
0	0.526	0.718	0.607	4017
1	0.537	0.335	0.413	3912
accuracy			0.529	7929
macro avg	0.531	0.527	0.510	7929
weighted avg	0.531	0.529	0.511	7929

It can be seen from the tables that “topic_specialization” has the strongest performance among all engineered variables. All other variables have weak predictive power for article popularity. However, other features may still be useful when combined with the full dataset, even if they show bad performance individually.

For the final step, we have created 2 final classification models that are built on all original numerical features and newly engineered ones. We have used Logistic Regression and Random Forest models to find the one that works better. We have evaluated the results using several metrics for binary classification: Accuracy, Precision, Recall, F1 and ROC-AUC. The results of the classification model predictions:

	Model	Best_params	Val_Accuracy	Test_Accuracy	Test_Precision	Test_Recall	Test_F1	Test_ROC_AUC
0	LogReg_Std	{'clf__C': 0.1, 'clf__penalty': 'l2'}	0.651	0.653	0.654	0.631	0.642	0.701
1	RandomForest	{'max_depth': 20, 'max_features': 0.5, 'n_esti...	0.667	0.662	0.654	0.669	0.662	0.725

Logistic Regression model predicts class 0 a little bit better than class 1.

Random Forest Model works generally better. It shows better results in every evaluation metric. It also balances class prediction. The accuracy of the model is 0.662 which is a moderately good, but not outstanding result. Such model accuracy numbers are explained by the nature of data: as we have seen previously it is hard to predict the popularity of a news article given the features that are in the dataset and the ones we have created. The resulting accuracy of 0.662 indicates that the model captures some patterns in the data that influence the target variable, although a big portion of the variance is not explained by the model.

After the final classification model we have performed the Feature Importance Analysis. As it can be seen on the graph (Figure 2) “kw_avg_avg” is the most important feature. We can see one of newly engineered features – “virality_score” as 8-th most important feature, which is a good result.

It is also important to mention that “topic_specialization” feature that we have created earlier showed the strongest individual predictive performance, while it was not included in the top 15 most important features. This happened because of the information overlap with existing variables. “topic_specialization” strongly correlates with other features, so the importance is “divided” between all of these features.

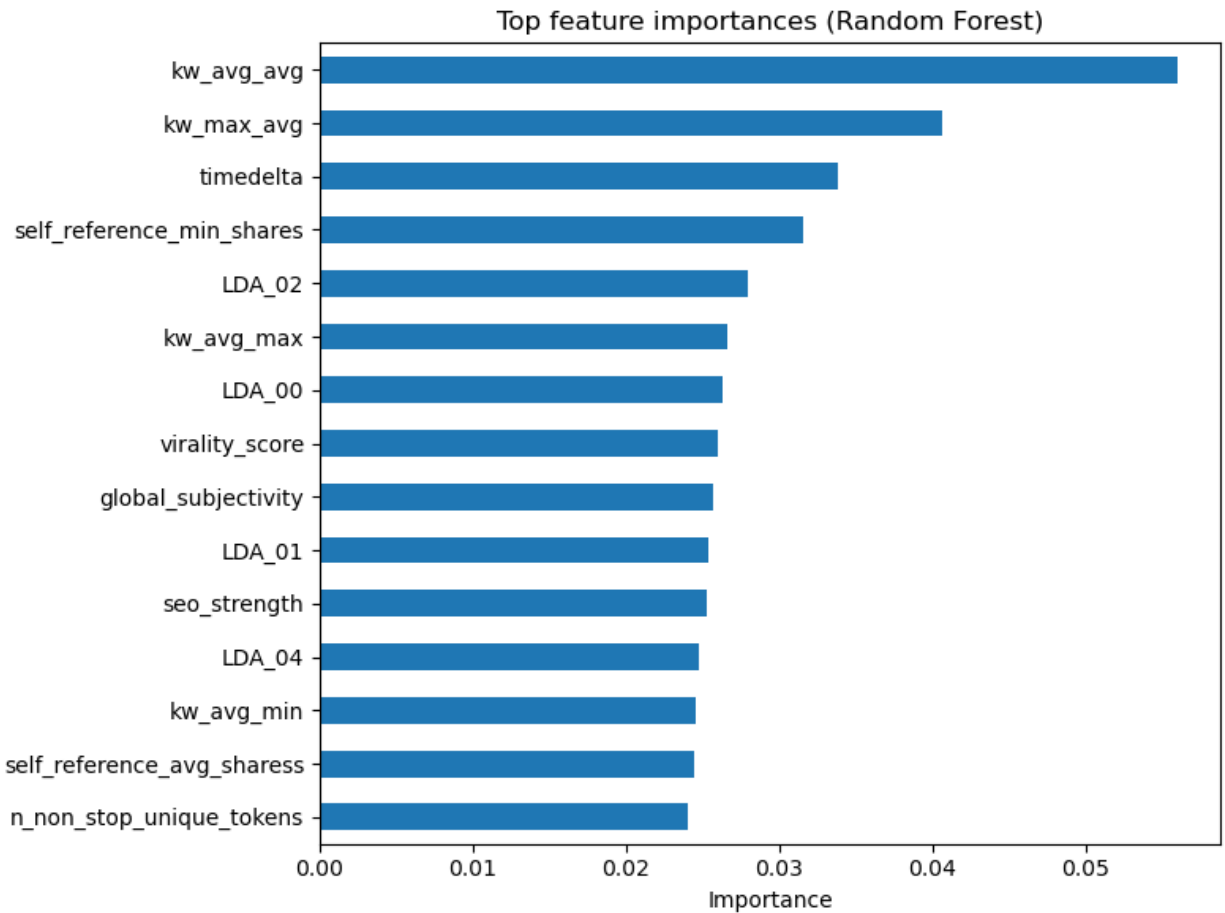


Figure 2: Feature Importance Analysis

CLUSTERING ANALYSIS:

For this stage of the project, we have performed the clustering of news articles on types of words, topics, general sentiment and newly engineered features. We have decided to use Agglomerative Clustering method here because it doesn't require assumptions about the shape of clusters (unlike K-Means) and is a faster method than DBSCAN. The result of splitting into 3 clusters:

```
cluster
0      38462
2       1181
1           1
Name: count, dtype: int64
```

As we can see cluster 0 contains most of the data points (typical article), cluster 2 is a smaller but also significant group of articles and cluster 3 is just one outlier.

To visualize the clusters, we have performed the Principal Component Analysis with 2 components (Figure 3).

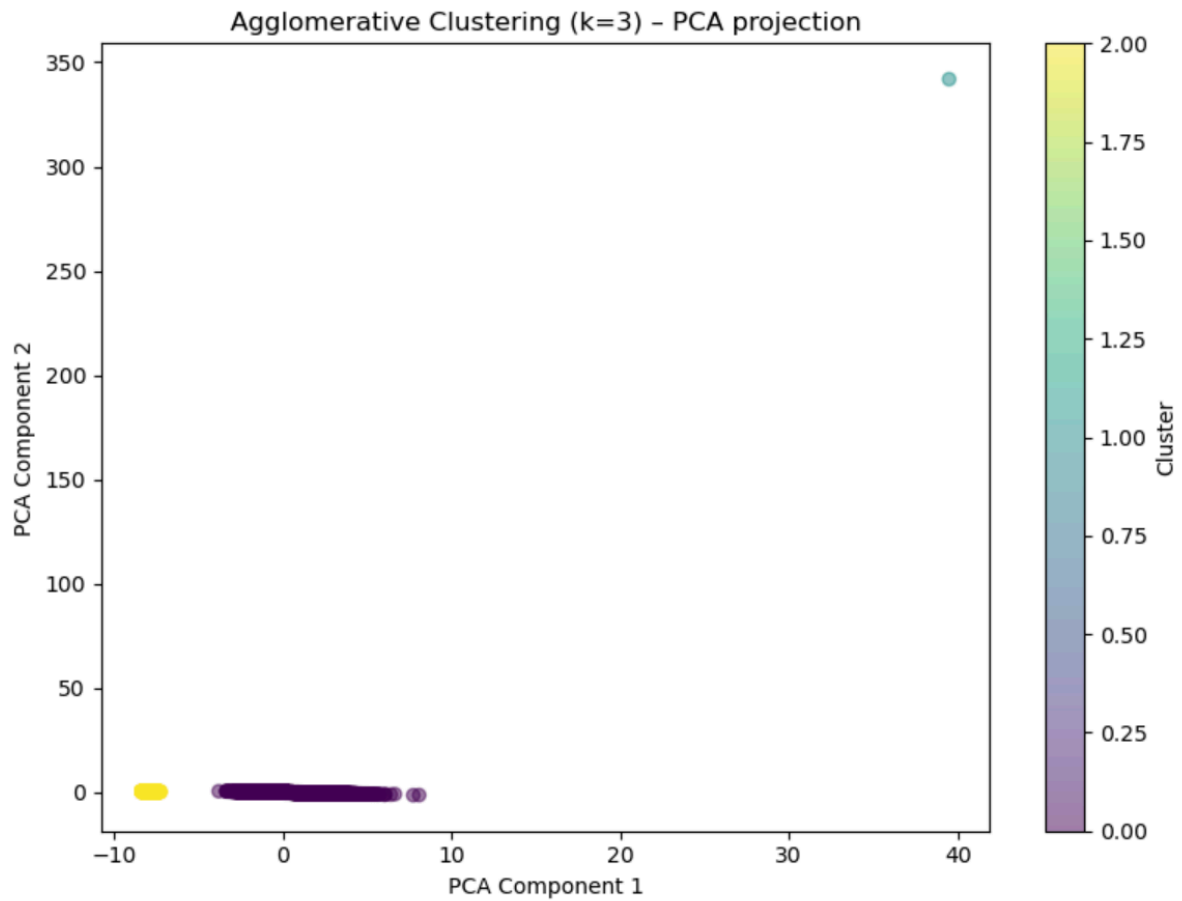


Figure 3: Agglomerative Clustering

We found that articles can be grouped into several natural clusters, which means that the popularity of articles is related not only to individual features, but also to their combinations that form different types of content. To get a better understanding of which groups of articles were data separated into, we have calculated the mean value of newly engineered features within each cluster:

	topic_specialization	sentiment_power	text_complexity	seo_strength	virality_score
cluster					
0	2513.820675	0.13225	2.345295	3.945685	0.705693
1	2970.487034	0.00000	2.630000	3.665726	1.217938
2	1227.025999	0.00000	0.000000	4.127784	-0.466366

The conclusion we can make here:

Cluster 0 – “Balanced, moderately popular articles”. They focus mainly on one topic (“topic_specialization” is high), the sentiment power is not dominant, text complexity is moderate, “virality_score” is positive, so the content has potentiality to be viral.

Cluster 2 – “Broad, SEO-oriented, but less viral articles”. They cover many topics or don’t go deep into the topic (“topic_specialization” is low) , there is no emotional component (sentiment_power = 0), the texts are simple (text_complexity = 0), “virality_score” is negative, so the content has low potentiality to be viral.

CONCLUSION:

In this project, we investigated whether textual, topical, and sentiment-related features of online news articles correspond with their popularity. To address this question, we combined feature engineering, supervised classification, feature importance analysis, and unsupervised clustering.

- The best-performing classification model, that we have created, achieved an accuracy of approximately 0.66, indicating a meaningful but moderate ability to distinguish popular from non-popular articles.
- In the Cluster Analysis we saw that articles with different feature combinations form stable clusters that influence article popularity differently. This confirms that article popularity cannot be explained by one feature. Type of content, rather than a single feature, plays a key role.

General conclusion is that it is hard to predict online news popularity. No single feature fully explains popularity, but the combination of supervised and unsupervised learning methods provides a better understanding of how different content types correlate with news article popularity.

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